TMDb Movie Data Analysis: Final Report

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1. Summary

This project analyzed 18 blockbuster movies from the TMDb API to measure financial performance (revenue, budget, profit, ROI), audience reception (ratings, votes, popularity), and the performance of franchises and directors. All 18 movies earned over $1.24 billion, together generating about $30.6 billion in revenue against $3.85 billion in production budgets.

2. Methodology

Extraction. Movie details and credits were requested from two TMDb endpoints using a Bearer token stored in a .env file. Of 19 requested IDs, 18 succeeded; ID 0 returned a 404 error because it is not a valid movie ID.

Cleaning. Irrelevant columns were dropped, nested JSON fields (collection, genres, languages, countries, companies) were flattened, data types were corrected, zero budgets and revenues were treated as missing, and values were converted to millions of USD. Duplicate, invalid, incomplete (fewer than 10 values), and unreleased records were filtered out. No rows were removed, giving a final dataset of 18 movies and 22 columns.

Analysis. Profit (revenue − budget) and ROI (revenue ÷ budget) were calculated. A reusable ranking function ranked movies by each KPI, with ROI limited to budgets of at least $10M and ratings to at least 10 votes. Movies were then grouped by collection and by director, and the findings were visualized with Matplotlib.

3. Key Insights

3.1 Revenue, Profit, and ROI

Movie	Budget ($M)	Revenue ($M)	Profit ($M)	ROI
Avengers: Endgame	356	2,927.4	2,571.4	8.22×
Avatar	237	2,923.7	2,686.7	12.34×
Titanic	200	2,264.2	2,064.2	11.32×
Star Wars: The Force Awakens	245	2,068.2	1,823.2	8.44×
Avengers: Infinity War	300	2,052.4	1,752.4	6.84×

Table 1. Top 5 movies by revenue

Avengers: Endgame earned the most revenue, but Avatar earned the highest profit and the highest ROI because it reached almost the same revenue with a $119M smaller budget. Titanic and Jurassic World (11.14×) also showed very high ROI. Star Wars: The Last Jedi had the lowest ROI (4.45×) and the lowest profit ($1,034.4M) despite a $300M budget.

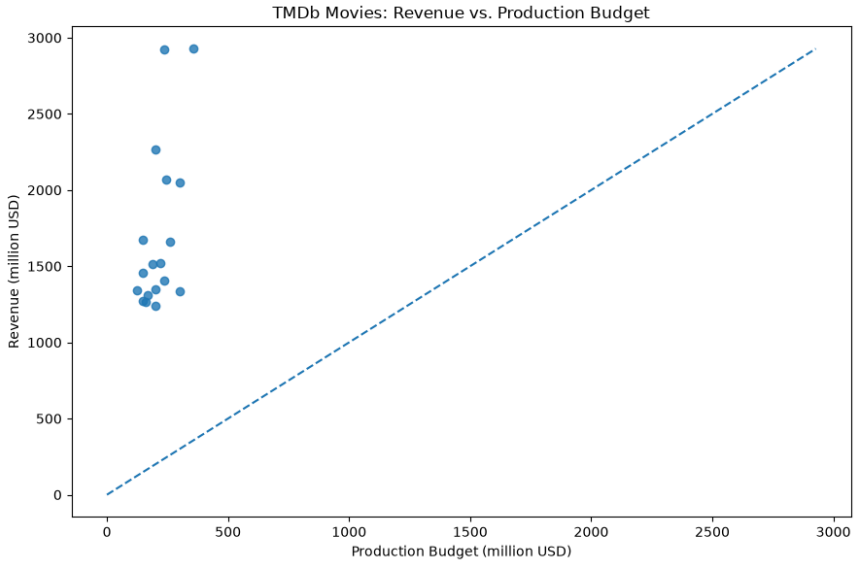


Figure 1. Revenue vs. production budget (dashed line: revenue = budget)

Every movie sits far above the break-even line. Budget and revenue show only a moderate positive relationship (r ≈ 0.58), so larger budgets did not guarantee proportionally larger returns.

3.2 Ratings and Popularity

Ratings ranged narrowly from 6.5 to 8.25. The Avengers films led in both quality and attention: Avengers: Infinity War had the highest rating (8.25) and was by far the most popular (142.7), followed by Endgame (8.20) and The Avengers (8.10), which also had the most votes (40,032). Box-office success did not always mean approval: both Jurassic World films were among the lowest rated (6.50 and 6.70).

3.3 Advanced Searches

The searches for Science Fiction/Action movies starring Bruce Willis and for Uma Thurman movies directed by Quentin Tarantino returned no results, because none of these films is among the 18 selected movies.

3.4 Franchises and Directors

Franchise	Movies	Total revenue ($M)	Mean rating
The Avengers Collection	4	7,904.1	7.96
Star Wars Collection	2	3,402.6	6.97
Jurassic Park Collection	2	2,982.0	6.60
Avatar Collection	1	2,923.7	7.61
Frozen Collection	2	2,727.9	7.24

Table 2. Top 5 franchises by total revenue

Director	Movies	Total revenue ($M)	Mean rating
James Cameron	2	5,187.9	7.76
Anthony Russo / Joe Russo	2	4,979.9	8.22
Joss Whedon	2	2,924.2	7.69
Jennifer Lee / Chris Buck	2	2,727.9	7.24
J.J. Abrams	1	2,068.2	7.20

Table 3. Top directors by total revenue (co-directors share credit)

The Avengers franchise produced about a quarter of all revenue in the dataset, more than double Star Wars. James Cameron led directors with just two films (Avatar and Titanic), while the Russo brothers had the highest mean rating (8.22).

3.5 Genre, Year, and Franchise Comparison

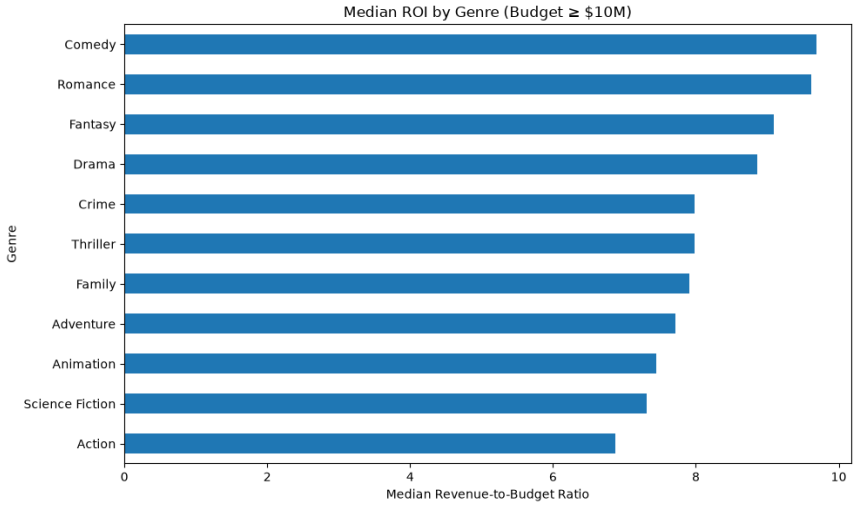


Figure 2. Median ROI by genre (budgets ≥ $10M)

Comedy (9.69×), Romance (9.62×), and Fantasy (9.09×) had the highest median ROI, while Action (6.87×) and Science Fiction (7.31×) had the lowest, suggesting that big action blockbusters earn the most but least efficiently. Some genres include only one or two movies, so these results are indicative.

By release year, 2015 had the highest combined revenue ($6,660.6M), but only because four of the selected movies were released that year. The two movies without a recorded collection (Titanic and Beauty and the Beast) had a higher median ROI (9.62× vs. 7.84×) than the 16 franchise movies, though a group of two is too small to generalize.

4. Conclusions

- **Scale and efficiency differ.** Endgame earned the most, but Avatar delivered the best profit and ROI.
- **Higher budgets show diminishing returns.** The most expensive films did not achieve the best ROI.
- **Franchises dominate.** 16 of 18 movies belong to a collection, led by the Avengers.
- **Revenue and ratings are related but not the same.** Marvel succeeded on both; Jurassic World earned well but was rated low.

5. Limitations and Future Work

The sample contains only 18 billion-dollar movies, so results cannot be generalized to the wider industry. Profit and ROI exclude marketing and distribution costs, revenues are not adjusted for inflation, and TMDb ratings reflect only TMDb users. Future work should use a larger dataset across all budget levels, adjust for inflation, and apply statistical tests before drawing firm conclusions.